

# The `natex-lualatex` package

Nathan G. Phillips

September 16, 2026

## Abstract

A collection of commands focused on consistent notation for mathematics, physics, and engineering. The repository for this package can be found at: <https://github.com/amilkyboi/natex>.

## Contents

|          |                                   |          |
|----------|-----------------------------------|----------|
| <b>1</b> | <b>Included packages</b>          | <b>2</b> |
| <b>2</b> | <b>Commands</b>                   | <b>3</b> |
| 2.1      | Automated bracing . . . . .       | 3        |
| 2.2      | Vector notation . . . . .         | 3        |
| 2.3      | Dirac notation . . . . .          | 4        |
| 2.4      | Set notation . . . . .            | 4        |
| 2.5      | Matrix notation . . . . .         | 5        |
| 2.6      | Linear operators . . . . .        | 5        |
| 2.7      | Probability . . . . .             | 5        |
| 2.8      | Trigonometric functions . . . . . | 6        |
| 2.9      | Mathematical constants . . . . .  | 6        |
| 2.10     | Other . . . . .                   | 6        |

## 1 Included packages

This package requires the `fontsetup`, `mathtools`, and `unicode-math` packages.

## 2 Commands

### 2.1 Automated bracing

| Command             | Usage                     | Output           | Definition         |
|---------------------|---------------------------|------------------|--------------------|
| <code>\abs</code>   | <code>\abs{x}</code>      | $ x $            | absolute value     |
| <code>\norm</code>  | <code>\norm{x}</code>     | $\ x\ _2$        | $\ell^2$ norm      |
| <code>\comm</code>  | <code>\comm{x}{y}</code>  | $[x, y]_-$       | commutator         |
| <code>\acomm</code> | <code>\acomm{x}{y}</code> | $[x, y]_+$       | anticommutator     |
| <code>\pb</code>    | <code>\pb{x}{y}</code>    | $\{x, y\}$       | Poisson bracket    |
| <code>\order</code> | <code>\order{x}</code>    | $\mathcal{O}(x)$ | order of magnitude |
| <code>\eval</code>  | <code>\eval{x}</code>     | $x $             | evaluation limits  |

### 2.2 Vector notation

| Command             | Usage                          | Output                                 | Definition       |
|---------------------|--------------------------------|----------------------------------------|------------------|
| <code>\vb</code>    | <code>\vb{x}</code>            | $\boldsymbol{x}$                       | bold vector      |
| <code>\vu</code>    | <code>\vu{x}</code>            | $\hat{\boldsymbol{x}}$                 | unit vector      |
| <code>\matx</code>  | <code>\matx{A}</code>          | $\boldsymbol{A}$                       | matrix           |
| <code>\tens</code>  | <code>\tens{T}</code>          | $\boldsymbol{T}$                       | tensor           |
| <code>\vdot</code>  | <code>\vb{x}\vdot\vb{y}</code> | $\boldsymbol{x} \cdot \boldsymbol{y}$  | dot product      |
| <code>\vcrs</code>  | <code>\vb{x}\vcrs\vb{y}</code> | $\boldsymbol{x} \times \boldsymbol{y}$ | cross product    |
| <code>\del</code>   | <code>\del</code>              | $\nabla$                               | del              |
| <code>\grad</code>  | <code>\grad{x}</code>          | $\nabla x$                             | gradient         |
| <code>\divr</code>  | <code>\divr{\vb{x}}</code>     | $\nabla \cdot \boldsymbol{x}$          | divergence       |
| <code>\curl</code>  | <code>\curl{\vb{x}}</code>     | $\nabla \times \boldsymbol{x}$         | curl             |
| <code>\slap</code>  | <code>\slap{x}</code>          | $\nabla^2 x$                           | scalar Laplacian |
| <code>\vlap</code>  | <code>\vlap{\vb{x}}</code>     | $\nabla^2 \boldsymbol{x}$              | vector Laplacian |
| <code>\dalem</code> | <code>\dalem</code>            | $\square$                              | d'Alembertian    |

### 2.3 Dirac notation

| Command            | Usage                       | Output                   | Definition             |
|--------------------|-----------------------------|--------------------------|------------------------|
| <code>\bra</code>  | <code>\bra{x}</code>        | $\langle x $             | bra                    |
| <code>\ket</code>  | <code>\ket{x}</code>        | $ x\rangle$              | ket                    |
| <code>\ev</code>   | <code>\ev{x}</code>         | $\langle x \rangle$      | expectation value      |
| <code>\ip</code>   | <code>\ip{x}{y}</code>      | $\langle x y\rangle$     | inner product          |
| <code>\op</code>   | <code>\op{x}{y}</code>      | $ x\rangle\langle y $    | outer product          |
| <code>\mel</code>  | <code>\mel{x}{y}{z}</code>  | $\langle x y z\rangle$   | matrix element         |
| <code>\rmel</code> | <code>\rmel{x}{y}{z}</code> | $\langle x  y  z\rangle$ | reduced matrix element |

### 2.4 Set notation

| Command           | Usage                             | Output            | Definition              |
|-------------------|-----------------------------------|-------------------|-------------------------|
| <code>\N</code>   | <code>\N</code>                   | $\mathbb{N}$      | set of natural numbers  |
| <code>\Z</code>   | <code>\Z</code>                   | $\mathbb{Z}$      | set of integers         |
| <code>\Q</code>   | <code>\Q</code>                   | $\mathbb{Q}$      | set of rational numbers |
| <code>\R</code>   | <code>\R</code>                   | $\mathbb{R}$      | set of real numbers     |
| <code>\C</code>   | <code>\C</code>                   | $\mathbb{C}$      | set of complex numbers  |
| <code>\set</code> | <code>\set{a, b, c}</code>        | $\{a, b, c\}$     | set notation            |
| <code>\set</code> | <code>\set{a \ given b, c}</code> | $\{a \mid b, c\}$ | set builder notation    |

## 2.5 Matrix notation

| Command             | Usage                                        | Output                                                        | Definition           |
|---------------------|----------------------------------------------|---------------------------------------------------------------|----------------------|
| <code>\pmx</code>   | <code>\pmx{1 &amp; 2 \\\ 3 &amp; 4}</code>   | $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$                | parenthetical matrix |
| <code>\bm{x}</code> | <code>\bm{x}{1 &amp; 2 \\\ 3 &amp; 4}</code> | $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$                | bracketed matrix     |
| <code>\vm{x}</code> | <code>\vm{x}{1 &amp; 2 \\\ 3 &amp; 4}</code> | $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$                | vertical matrix      |
| <code>\cm{x}</code> | <code>\cm{x}{1 &amp; 2 \\\ 3 &amp; 4}</code> | $\left\{ \begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix} \right\}$ | curly matrix         |
| <code>\tr</code>    | <code>\tr{\mat{x}{A}}</code>                 | $\text{tr } A$                                                | trace                |
| <code>\adj</code>   | <code>\adj{\mat{x}{A}}</code>                | $\text{adj } A$                                               | adjugate             |
| <code>\tp</code>    | <code>\mat{x}{A}^{\sim\text{tp}}</code>      | $A^{\text{T}}$                                                | transpose            |
| <code>\cc</code>    | <code>\mat{x}{A}^{\sim\text{cc}}</code>      | $A^*$                                                         | complex conjugate    |
| <code>\hc</code>    | <code>\mat{x}{A}^{\sim\text{hc}}</code>      | $A^{\dagger}$                                                 | Hermitian conjugate  |

## 2.6 Linear operators

| Command           | Usage                | Output    | Definition      |
|-------------------|----------------------|-----------|-----------------|
| <code>\sop</code> | <code>\sop{x}</code> | $\hat{x}$ | scalar operator |
| <code>\vop</code> | <code>\vop{x}</code> | $\hat{x}$ | vector operator |

## 2.7 Probability

| Command            | Usage                 | Output           | Definition                   |
|--------------------|-----------------------|------------------|------------------------------|
| <code>\erf</code>  | <code>\erf{x}</code>  | $\text{erf } x$  | error function               |
| <code>\erfc</code> | <code>\erfc{x}</code> | $\text{erfc } x$ | complementary error function |

## 2.8 Trigonometric functions

| Command              | Usage                   | Output                    | Definition          |
|----------------------|-------------------------|---------------------------|---------------------|
| <code>\asin</code>   | <code>\asin{x}</code>   | $\operatorname{asin} x$   | arcsine             |
| <code>\acos</code>   | <code>\acos{x}</code>   | $\operatorname{acos} x$   | arccosine           |
| <code>\atan</code>   | <code>\atan{x}</code>   | $\operatorname{atan} x$   | arctangent          |
| <code>\asec</code>   | <code>\asec{x}</code>   | $\operatorname{asec} x$   | arcsecant           |
| <code>\arcsec</code> | <code>\arcsec{x}</code> | $\operatorname{arcsec} x$ | arcsecant           |
| <code>\acsc</code>   | <code>\acsc{x}</code>   | $\operatorname{acsc} x$   | arccosecant         |
| <code>\arccsc</code> | <code>\arccsc{x}</code> | $\operatorname{arccsc} x$ | arccosecant         |
| <code>\acot</code>   | <code>\acot{x}</code>   | $\operatorname{acot} x$   | arccotangent        |
| <code>\arccot</code> | <code>\arccot{x}</code> | $\operatorname{arccot} x$ | arccotangent        |
| <code>\sech</code>   | <code>\sech{x}</code>   | $\operatorname{sech} x$   | hyperbolic secant   |
| <code>\csch</code>   | <code>\csch{x}</code>   | $\operatorname{csch} x$   | hyperbolic cosecant |

## 2.9 Mathematical constants

| Command           | Usage             | Output | Definition     |
|-------------------|-------------------|--------|----------------|
| <code>\img</code> | <code>\img</code> | $i$    | imaginary unit |
| <code>\eul</code> | <code>\eul</code> | $e$    | Euler's number |
| <code>\pie</code> | <code>\pie</code> | $\pi$  | pi             |

## 2.10 Other

| Command             | Usage               | Output              | Definition     |
|---------------------|---------------------|---------------------|----------------|
| <code>\real</code>  | <code>\real</code>  | $\operatorname{Re}$ | real part      |
| <code>\imag</code>  | <code>\imag</code>  | $\operatorname{Im}$ | imaginary part |
| <code>\defas</code> | <code>\defas</code> | $\coloneqq$         | defined as     |